

Evoltrya OS — Document 3: Roadmap

What This Document Is

Document 1 laid out the exhaustive functional scope. Document 2 fixed the architecture, the design principles, the module map, and the precise current-state positioning. This Document 3 turns that blueprint into an executable build plan — phased, sequenced, and anchored to milestones. It is the bridge from "what the system is" to "in what order it gets built."

This roadmap is written for two readers at once: the product lead deciding what to build next and why, and the execution layer translating each phase into database schema, server logic, and UI. Each phase therefore states its goal and business rationale (the strategic layer) alongside its concrete deliverables, dependencies, and definition of done (the execution layer). It stops short of field-level table design — that belongs to each build session — but it is specific about what gets built, in what order, and what "complete" means.

The Sequencing Logic

The build order is set by the module rollout priority established in Document 1:

1. Manufacturing (Processing) -> 2. Warehousing (Inventory/Logistics) -> 3. Finance -> 4. Procurement + Sales

This priority deliberately departs from the natural data-flow order. Materials flow through the business as procurement -> inbound -> processing -> inventory -> sales — buy, receive, transform, store, sell. Yet the build priority puts processing first and procurement last. This is a sound commercial judgement, and the roadmap is built around it rather than around the data flow, for three reasons:

- **Processing is the operational heart and the sharpest pain.** One-to-many conversion, batch-level COGS, and metal recovery rate are the capabilities the business most needs and the ones Xero most conspicuously cannot provide. Rebuilding processing first delivers the system's distinctive value immediately.
- **The inbound module already records batches.** Processing does not have to wait for a full procurement module — it can draw on the existing inbound batches as its input source today. The full procurement cycle (requisition -> PO -> approval -> three-way matching) can be layered on later.
- **Value compounds downstream.** Once processing produces true batch costs and inventory holds accurate multi-dimensional stock, finance can compute real per-batch margin, and procurement/sales can price against metal content. Each phase makes the next more valuable.

How the roadmap reconciles priority with data flow. Each phase is sequenced by the priority above, but every phase states how it connects to what already exists. Processing is built first against the

current inbound batches; when procurement is built later, inbound is "upgraded" from a standalone record into the result of a purchase-order goods receipt — without disturbing the processing module that already consumes it. This is the same "reserve the interface, build what is needed now" principle that runs through the whole system: build the high-value core first, and let the data flow close up around it as later phases land.

Compliance is woven in, not stacked. Compliance is the industry-core concern, but it is not a single late phase. It is the gate that stands before each business action — hazardous-waste handling before processing, import permits before procurement, cross-border transfer documents before inter-entity movement, export compliance before sales. Rather than building one large compliance module at the end, each phase brings out the compliance surface it needs, and a consolidating pass late in the roadmap unifies them into the regulation library and filing engine. This suits compliance's real nature: a validation checkpoint embedded in the operating workflow, not a separate ledger.

The permission system is a unified final pass. Its interface has been reserved from the start — every table carries the ownership dimension and RLS hooks. Rather than implementing permissions piecemeal per module, the roadmap implements them once, near the end, when the modules they govern are built and their access patterns are known.

Three things run through every phase: master-data enrichment (suppliers, customers, materials growing from basic fields toward full management), cross-cutting capabilities (audit, coding, notifications maturing as modules need them), and HR data integration (kept light, its execution outsourced, its data integrated where it touches cost and compliance).

Phase 0 — Foundation and Show Units (Complete)

This is the baseline the roadmap plans forward from. It is already built and in production, and it is stated here so the plan begins from an honest starting line.

What stands:

- **Framework foundations** — authentication and RLS (hardened), bilingual i18n with build-time parity, the atomic database transaction pattern, soft deletes, automatic coding, updated_ at triggers, and a module-development template proven reusable across both master-data and transaction modules.
- **Master-data modules at feature-complete list level** — suppliers (with 8-state machine, compliance sub-table, attachments), customers, and materials (with category filter and attachments). Full URL-driven search/filter/sort/pagination, CSV export, and private-bucket attachments.
- **Transaction modules at list level** — inbound and output, each with cross-table search (by code and related party name), stage/state filters, and related-party dropdown filters.

What this baseline gives the roadmap: a proven construction method and a validated template. Every phase below is executed by applying this method — establish database conventions and SQL, review before writing, build the UI in verified steps — to a new slice of business functionality. The remaining

work is not invention; it is disciplined application of a proven pattern to progressively deeper business logic.

Milestone 0 (met): a user can manage master data and record inbound/output batches through production-quality list interfaces, in two languages, with attachments and export.

Phase 1 — Processing Reborn (Highest Priority)

Goal. Transform the processing module from a skeleton that logs a material balance into a production-management engine built around one-to-many recycling conversion, with true batch-level cost. This is the phase that delivers the system's distinctive value — the capabilities Xero structurally cannot provide.

Business rationale. Processing is the operational heart. Until the system can answer "how much black mass did this battery batch yield, at what recovery rate, at what cost per output," it records history without explaining it. This phase makes the traceability chain acquire its cost dimension.

What gets built:

- **One-to-many conversion, natively.** A single input batch (or several) yields multiple distinct output materials — black mass, copper, aluminium, plastics, residue — with quantity conservation enforced at the database layer. The existing input/output-leg structure and atomic commit/rollback are the foundation; this phase builds production management on top.
- **Work order and execution.** A production plan / work order (what to process, how much, when), with an actual-execution record against plan. The work order becomes the unit that carries cost.
- **Yield and recovery analysis.** Loss-rate analysis with trend and anomaly detection, and metal recovery rate (input metal vs output metal) — the core industry KPI. Conservation is enforced not only on weight but toward metal-element accounting.
- **Batch COGS with configurable allocation.** Processing cost — raw material plus labour, energy, equipment depreciation, and consumables — apportioned to each output batch. The allocation basis (by weight, by metal value, or by market value) is an explicit, configurable choice, because it directly determines each output's reported gross margin.
- **Single-stage now, multi-operation reserved.** The single-stage model is implemented; the data structure is shaped so that a future "operation" layer (dismantle -> crush -> pyrolysis -> hydrometallurgy) can be inserted beneath the work order as an extension, not a rewrite.

Compliance surface brought out this phase: hazardous-waste handling context on the processing floor — the operational audit trail for who processed, weighed, and released each batch (a first concrete use of the operation-log foundation).

Connection to the data flow: processing draws its inputs from the existing inbound batches. No procurement module is required yet. Outputs feed the read-only inventory balance today and the full inventory ledger in Phase 2.

Dependencies: none beyond Phase 0. The inbound batches and the atomic commit/rollback RPCs already exist.

Definition of done (Milestone 1): a processing run converts one or more input batches into multiple output batches; the system reports yield percentage, metal recovery rate, and a per-output-batch cost on a chosen allocation basis; conservation and atomicity are enforced at the database layer; and the run carries a full operational audit trail.

Phase 2 — Warehousing and Inventory

Goal. Turn the read-only material balance into a real-time, multi-dimensional stock ledger that the business can trust for accuracy — the most-cited inventory pain — and that values stock by metal content, not weight alone.

Business rationale. Processing (Phase 1) now produces multiple output batches; those batches must live somewhere accurate. Accuracy is the stated inventory pain, and valuation by metal content is what makes inventory value reflect true commercial worth. This phase is what lets a stocktake reconcile and an auditor trace physical stock to the books.

What gets built:

- **Real-time multi-dimensional stock ledger.** Stock by material, batch, bin, and entity, with a movement ledger where every in/out is a traceable line. Status-differentiated stock: available, in-inspection, in-transit, frozen — the multi-status accuracy that is the core challenge.
- **Batch and aging management.** Batch shelf-life and age tracking (hazardous waste may have storage time limits), with aging-stock alerts. FIFO or specified-batch issue.
- **Stocktake.** A yearly full stocktake with variance handling (adjustment and approval of book-vs-physical mismatch) and stocktake history. Cycle/spot counts supported for interim checks.
- **Metal-content-based valuation.** Stock valued on a moving weighted-average basis, with valuation linked to assay results, metal percentage, moisture, and impurity — so two equal-weight batches of black mass can hold different values.
- **Alerts.** Safety-stock alerts (below reorder point prompts procurement) and slow-moving/overstock alerts.

Warehousing (logistics) built alongside: warehouse/bin master data (managed down to bin level, per the decision), stock-in/stock-out documents folding the current inbound/output into a complete in/out system, and packaging/labelling with scan-based stock in/out where feasible (weighbridge entry, batch QR).

Compliance surface brought out this phase: hazardous-waste zone/storage requirements (focused vs non-focused material handling), and the batch-level compliance attributes (soaked/voltage status, repurpose vs recycle classification) that inventory must carry as structured data.

Connection to the data flow: inventory receives from processing (output put-away) and inbound (receipt), and issues to sales (Phase 4). Moving weighted-average valuation feeds finance (Phase 3).

Dependencies: Phase 1 (processing outputs are the primary new stock-in). The inbound/output records from Phase 0 fold in here.

Definition of done (Milestone 2): stock is visible in real time by material/batch/bin/entity across differentiated statuses; a full stocktake can be performed and reconciled; stock is valued on a moving weighted-average basis linked to metal content; and aging and safety-stock alerts fire.

Phase 3 — Finance Core

Goal. Build the finance module as the operations-and-cost engine that Xero cannot be — anchored to the fifteen-point limitation study — with batch-level cost accounting and business-native journal automation at its centre.

Business rationale. With processing producing batch costs (Phase 1) and inventory holding valued stock (Phase 2), finance can now do what motivated the whole system: compute the true gross margin of each batch and each production run, and generate accounting from business data instead of re-keying it. This is the phase that upgrades the OS from a business system to a management system.

What gets built:

- **Batch-level cost accounting (the priority).** Batch cost and product cost — raw material plus processing plus apportionment — rolled up from Phases 1 and 2. Standard cost vs actual. True per-batch and per-work-order gross margin. This is the number the business most needs and Xero structurally cannot produce, so it leads the phase.
- **Business-native journal automation.** Business documents auto-generate journals: a purchase generates a payable journal, a sale a receivable journal, stock movements an inventory journal — exploiting business data being native to the system. This is the structural answer to Xero's central weakness.
- **General ledger.** Chart of accounts (two sets, SG/EU), manual plus auto-generated journals, period management (month-end, year-end, period lock), multiple ledgers across entities.
- **AR/AP.** Accounts payable (purchase invoices, payment clearing, ageing) and accounts receivable (sales invoices, collection clearing, ageing, chasing), with statement reconciliation.
- **Treasury.** Bank accounts and statements, cash flow (actual and forecast), payment plans and approvals — supporting the payment-management model (deposit / arrival / post-assay staging).
- **Tax and multi-currency.** SG GST and EU VAT handled per regime; business transactions in USD with translation; intra-EU VAT rules where relevant.
- **Reporting.** The three statements, multi-entity consolidated statements (the two independent P&L entities pulled together for group analysis), management reports, budget vs actual.

Compliance surface brought out this phase: the physical-to-financial linkage an auditor needs — the chain from GRN to pallet, work order, output batch, goods issue, sales invoice, and general ledger — and period lock as an audit control.

Connection to the data flow: finance sits downstream of nearly every module — procurement payables, sales receivables, processing/inventory cost, logistics expense — and performs multi-entity consolidation.

Dependencies: Phases 1 and 2 (cost and valuation are the inputs to cost accounting). Sales-side receivables mature further in Phase 4.

Definition of done (Milestone 3): the system computes true per-batch and per-work-order gross margin; business documents auto-generate journals; the three statements and a multi-entity consolidation can be produced; and an auditor can trace a general-ledger figure back to the physical stock movement that produced it.

Phase 4 — Procurement and Sales

Goal. Close the two ends of the material lifecycle — the full procurement cycle at the front and the full sales cycle at the back — both driven by a metal-content pricing engine, so the system spans requisition to payment and inquiry to collection.

Business rationale. Processing, inventory, and finance (Phases 1-3) form the value-creating core. Procurement and sales are the market-facing ends that feed and drain it. They come after the core because the inbound module already covers basic intake and output already carries a customer field — the ends can wait while the high-value middle is built. Now they are completed and connected.

What gets built — Procurement:

- **Full cycle.** Purchase requisition -> approval -> purchase order -> supplier confirmation -> goods receipt -> inspection -> put-away -> invoice matching -> payment. Two-level approval (requester raises, supervisor approves; above 10k escalates to CFO).
- **Inbound upgraded.** The standalone inbound batch becomes the result of a PO goods receipt, with the ability to differentiate discrepancies between ordered and received — while remaining compatible with the processing module that already consumes inbound batches.
- **Metal-content pricing and two-stage settlement.** Raw material priced by assay (nickel-cobalt content x LME/SMM), received at an estimated price and adjusted after assay, with the payment-management model (deposit 80% / arrival 10% / post-assay 10%). Three-way matching (PO + GR + invoice).

What gets built — Sales:

- **Full cycle.** Inquiry/quote -> sales order -> approval -> shipment -> invoicing -> collection. Approval before shipment. Saleable-stock check and reservation against inventory.
- **Multiple pricing models, integrated.** Output priced against metal markets (LME/SMM), with multiple pricing models coexisting and selected per transaction or contract — spot and long-contract both supported, per contract terms. This is where the "different pricing models integrated into one module" requirement lands.
- **Product-focused analytics.** Sales performance and margin analysis (price minus the true cost from Phase 3), by product.

Pricing engine (shared). A configurable pricing engine, integrated into both procurement and sales, reads assay data and live metal-market quotes rather than hard-coding fixed prices. This is the single capability that most distinguishes the system's front and back ends from a generic ERP.

Compliance surface brought out this phase: import compliance and permits on procurement; export compliance on sales; supplier/customer qualification verification gating transactions.

Connection to the data flow: procurement feeds inbound/inventory and creates payables; sales draws from inventory, ships via logistics, and creates receivables. Both close the loop around the Phase 1-3 core.

Dependencies: Phase 3 (true cost is needed for real sales margin; payables/receivables post to the GL). Inventory reservation depends on Phase 2.

Definition of done (Milestone 4): a purchase runs from requisition through metal-content-priced, assay-adjusted settlement; a sale runs from quote through metal-priced invoicing and collection with an inventory reservation; and both post automatically to finance.

Phase 5 — Compliance Consolidation and Trade

Goal. Unify the compliance surfaces surfaced piecemeal across Phases 1-4 into a coherent compliance-and-trade module — the industry-core capability — turning scattered checkpoints into a regulation library, a cross-border transfer engine, and a filing system.

Business rationale. Compliance was woven into each earlier phase as the gate before each action. By this point the operational hooks exist (hazardous-waste handling, storage requirements, import/export permits, batch compliance attributes). This phase consolidates them so compliance becomes structured operational data that gates transactions — validating that a shipment, batch, or sale has its required documents before it proceeds — rather than loose attachments.

What gets built:

- **Qualification and licence management.** Company qualifications (hazardous-waste handling licence, import/export rights, permits, with expiry alerts), supplier/customer qualification verification, material compliance classification (which are hazardous waste, with control codes). The supplier compliance sub-table from Phase 0 generalises here.
- **Cross-border transfer compliance.** Basel-related documents (prior informed consent for cross-border hazardous-waste transfer), transfer application/approval management, full audit trail of transfer documents — supporting the SG/EU inter-entity material movement.
- **Import/export customs.** Customs declaration management, HS codes and duties, freight-forwarder/broker integration, document management (B/L, invoice, packing list, certificate of origin).
- **Regulatory filing.** Filing per jurisdiction, with filing data auto-generated from business data (the traceability chain feeds the filing forms) — the payoff of building on native business data.
- **Regulation library and alerts.** Applicable-regulation list (Basel, EU Battery Regulation, SG/EU local hazardous-waste regulations), update reminders, and a per-transaction compliance checklist.

Connection to the data flow: compliance touches procurement (import-material compliance), sales (export-product compliance), logistics (cross-border transport), suppliers/customers (qualifications), and traceability (the filing-data source).

Dependencies: the operational compliance hooks from Phases 1-4. Structured compliance data draws on the full traceability chain now in place.

Definition of done (Milestone 5): compliance documents and qualifications are managed with expiry alerts; a cross-border transfer can be applied for and tracked with a full document trail; filings are auto-generated from business data; and a transaction can be gated on a compliance checklist.

Phase 6 — Auxiliary Modules and Consolidation

Goal. Build the experience-and-efficiency layer that ties the operational modules together — the workbench, notifications, and reporting that make the system usable at a glance and across modules.

Business rationale. With the operational and financial core complete, the auxiliary modules multiply its value: a role-tailored home page surfaces what each user needs, notifications drive action, and a reporting centre answers cross-module questions. The tasks module (already full-featured from Phase 0) gains its business-module linkage here.

What gets built:

- **Dashboard/Workbench.** A personal workbench (my tasks, my documents, pending my approval), an operating dashboard (stock levels, procurement/sales/processing overview, finance overview), a role-tailored home page, and KPI visualisation (recovery rate, loss, turnover, margin).
- **Notification Center.** System notifications (approvals, alerts, expiries, exceptions), multi-channel delivery, and notification preferences.
- **Reporting Center.** Cross-module reports, custom reports/queries, scheduled reports and export — beyond each module's built-in exports.
- **Tasks linkage.** The tasks module connects to business modules — "this PO awaits your approval" auto-becomes a task — activating the linkage its data structure has reserved since Phase 0.
- **Industry News (optional).** Metal-quote and policy aggregation, feeding the pricing engine where useful.

Dependencies: the modules these surfaces summarise (Phases 1-5). Notifications and dashboards are only as useful as the data behind them.

Definition of done (Milestone 6): each role sees a tailored home page with live KPIs; notifications drive pending actions; and cross-module reports can be built and scheduled.

Final Phase — Unified Permission System

Goal. Implement, in a single coordinated pass, the module-level permission system whose interface has been reserved since the very first table.

Business rationale. Permissions were deliberately deferred, not forgotten. Every table has carried the ownership dimension and RLS hooks from the start, and the user-role structure was established early. Implementing permissions once — when all governed modules exist and their access patterns are known — is far cleaner than retrofitting each module piecemeal. This is the "reserve the interface, build it once at the end" principle at its largest scale.

What gets built:

- **Module-level access control.** The ten pre-defined roles (Super Admin/Owner, Procurement, Warehouse/Logistics, Processing/Production, Sales, Finance, Compliance, HR, Business Development, Read-only/Auditor) mapped to the modules each may enter.
- **Ownership-based data scope.** Data filtered by the department/function ownership dimension every record has carried, and by entity (SG/EU), activated through the RLS hooks already in place.
- **Users and identity.** User account management, user-to-entity association, and login methods (email/password plus Google/Microsoft SSO).
- **Approval workflows activated.** The value-bearing-action approvals designed into procurement, sales, and finance are bound to the role structure — requester to supervisor, above-threshold to CFO — with approval delegation and audit trail.

Dependencies: all modules whose access it governs (Phases 1-6). This phase is last by design.

Definition of done (Final Milestone): each user, by role and entity, sees and edits only what their permissions allow; the ten roles map to module access; data scope filters by ownership and entity; and value-bearing actions route through the designed approval chains. The system is now multi-user and permission-segregated — the last of the four foundations, fully realised.

Running Through Every Phase

Three streams of work do not occupy a phase of their own; they mature continuously alongside the phased build.

Master-data enrichment. Suppliers, customers, and materials grow from their current basic-fields-plus-list-features state toward full management, phase by phase, as the modules that consume them come online: supplier classification, rating, and the supplier audit process; customer credit management and contracts; material specifications, multi-level classification, and unit conversions (kg/tonne). Each enrichment lands when the business module that needs it is built — material specifications with processing, customer credit with sales, supplier audit with procurement.

Cross-cutting capabilities. Audit (operational and accounting), the coding system (extended to each new module with entity identifiers), and the module template itself mature as modules need them. The

attachment, export, and URL-driven-list capabilities already standard from Phase 0 are applied to every new module by default.

HR data integration. HR is kept deliberately light: execution is outsourced, and the system retains the integrated data — employee master data, the organisation structure that also feeds the permission system, and the compensation data that posts to finance as cost. It must be compliant with local (currently Singapore) requirements, and its end results are provided as raw data. HR touches the system where it matters — labour cost into processing, org structure into permissions, certificates into compliance — without building a full HR-execution engine.

Roadmap at a Glance

Phase	Focus	Priority basis	Key milestone
0 (done)	Foundation + show units	Baseline	Master data + inbound/output at production-quality list level
1	Processing reborn	Priority #1 (manufacturing)	One-to-many conversion, yield/recovery, batch COGS
2	Warehousing + inventory	Priority #2 (warehousing)	Real-time multi-dim ledger, stocktake, metal-content valuation
3	Finance core	Priority #3 (finance)	Batch margin, journal automation, multi-entity consolidation
4	Procurement + sales	Priority #4	Full cycles both ends, metal-content pricing engine
5	Compliance consolidation	Woven-in, then unified	Regulation library, cross-border transfer, auto-filing
6	Auxiliary + consolidation	Experience layer	Role dashboards, notifications, cross-module reporting
Final	Unified permissions	Reserved from start	Module-level RBAC, ownership scope, approval chains

Guiding Principles for Execution

The roadmap is sequenced, but it is not rigid. A few principles govern how it is walked:

- **Value first, flow second.** The order follows business value (processing first), letting the data flow close up around the high-value core as later phases land. This is why procurement, the flow's start, comes near the end.

- **Build on native data.** Each phase exploits that business data is native to the system — the reason journal automation, auto-filing, and metal-content valuation are achievable rather than aspirational.
- **Reserve, then build once.** Interfaces are reserved early (permissions, multi-operation processing, task collaboration) and implemented once at the right time, avoiding piecemeal retrofit.
- **One slice at a time, verified.** Each phase is built the way Phase 0 was — database conventions and SQL first, reviewed before writing, UI in verified steps — applying a proven method to progressively deeper business logic.
- **Compliance as a gate, not a ledger.** Compliance is validated inline at each business action and consolidated once, matching its nature as an operating-workflow checkpoint.

From here, each phase becomes a series of build sessions, each session a slice of one module, each slice carried by the same method that produced the foundation. The blueprint set the destination; this roadmap sets the order of the journey.